

- 11 (a) (i) frequency of carrier wave varies M1
in synchrony with displacement of information signal A1 [2]
- (ii) 1. zero (accept constant) B1 [1]
2. upper limit 530 kHz B1
lower limit 470 kHz B1
changes upper limit → lower limit → upper limit at 8000 s⁻¹ B1 [3]
- (b) e.g. more radio stations required / shorter range
more complex electronics
larger bandwidth required
(any two sensible suggestions, 1 each) B2 [2]
- 12 (a) (i) picking up of signal in one cable M1